# CPSC 250 - Programming for Data Manipulation
# Final Exam

## Instructions

You have 150 minutes. This is a closed-book, in-class exam. No access to the internet or external software is allowed. Show all work clearly and legibly.

# Part I: Multiple Choice (30 points, 2 points each)

Circle the best answer.

1. What is the output of the following code? `a = [1, 2, 3]`
   ```
   b = a
   b[0] = 99
   print(a)
   ```
   A. `[1, 2, 3]`
   B. `[99, 2, 3]`
   C. `[1, 99, 3]`
   D. Error

2. What does `df.dropna()` do in pandas?
   A. Fills in missing values
   B. Drops rows or columns with missing values
   C. Drops duplicate rows
   D. Drops specified columns

3. What is the purpose of the `__str__` method in a class?
   A. Initialize the object
   B. Represent the object as a string
   C. Destroy the object
   D. Compare two objects

4. What is the primary benefit of inheritance in object-oriented programming?
   A. Improved speed
   B. Data hiding
   C. Code reuse
   D. Loop simplification

5. In a recursive function, what is the base case?
   A. The case where an exception is raised
   B. The function that calls itself
   C. The condition where recursion stops
   D. The loop inside the function

6. Which of the following is true about lists and tuples?
    A. Lists are immutable, tuples are mutable
    B. Both are mutable
    C. Both are immutable
    D. Lists are mutable, tuples are immutable
7. What does the `curve_fit` function from `scipy.optimize` do?
    A. Plots data using a curve
    B. Fits data to a specified function
    C. Fits a line to data using OLS
    D. Returns the correlation coefficient
8. What is the time complexity of searching in a balanced binary search tree?
    A. `O(1)`
    B. `O(n)`
    C. `O(log n)`
    D. `O(n log n)`
9. Which of the following statements creates a dictionary?
    A. `d = {}`
    B. `d = []`
    C. `d = ()`
    D. `d = set()`
10. What happens when you override the `__eq__` method in a class?
    A. You redefine how two instances are compared with ==
    B. You make your class hashable
    C. You initialize the object
    D. You compare types of objects
11. What does the pandas method `read_csv()` return?
    A. A list of strings
    B. A NumPy array
    C. A Series
    D. A DataFrame

12. What does operator overloading allow in Python?

    A. Redefining loops
    B. Creating multiple classes in one file
    C. Using operators like + with user-defined types
    D. Automatically generating documentation

13. In pandas, what does `df['column'].mean()` compute?

    A. The median
    B. The total
    C. The standard deviation
    D. The average

14. What is polymorphism in OOP?

    A. The ability of different classes to share the same attributes
    B. The ability of objects to take on multiple forms via shared methods
    C. Hiding data inside an object
    D. Changing variable types at runtime

15. What is the correct way to inherit from a class in Python?

    A. `class A inherits B:`
    B. `class A: inherits B`
    C. `class A(B):`
    D. `class A<B>:`

## Part II: Error Identification (15 points)

The following code is meant to reverse a list recursively. Identify and correct all errors.

```
def reverse_list(lst):
    if len(lst) = 0:
        return []
    else:
        return reverse_list(lst[1:]) + lst[0]
```

# Part III: Code Writing (30 points)

**Q1. (6 points)** Write a function `is_palindrome(s)` that returns `True` if the string `s` is a palindrome, and `False` otherwise.

**Q2. (8 points)** Write a class `Student` with fields for `name` and `gpa`, and a `__str__` method that prints the student's name and GPA. Also include an `__lt__` method that compares students by GPA.

**Q3. (8 points)** Write a function `load_and_filter(filename)` that reads a CSV file into a pandas DataFrame, and returns only the rows where the value in the column `"score"` is greater than 80.

**Q4. (8 points)** Write a base class `Animal` with a method `make_sound()` that returns the string `"..."`. Create two derived classes `Dog` and `Cat` that override `make_sound()` to return `"woof"` and `"meow"`, respectively. Show how polymorphism allows you to loop over a list of animals and call `make_sound()` on each.

## Part IV: Code Comprehension and Commenting (25 points)

This function is designed to perform curve fitting using `curve_fit`. Comment each line and explain what the code is doing overall.

```
import numpy as np
from scipy.optimize import curve_fit

def model(x, a, b):
    return a * np.exp(b * x)

def fit_data(x, y):
    popt, pcov = curve_fit(model, x, y)
    print("a =", popt[0])
    print("b =", popt[1])
```

**What do `popt` and `pcov` represent in this context?**